

Intelligent Battery Chargers

Rugged Multi-Chemistry Charging for Industrial & Mission-Critical Systems



Precision High-Power Control.
Stable Programmable DC Output.
Built for Advanced Applications.

CRYONANO Intelligent Battery Chargers are high-reliability power systems engineered to deliver safe, stable, and efficient charging across a wide range of battery chemistries and operating environments.

Designed for embedded systems, off-grid installations, and mission-critical platforms, these chargers provide precise charging control, advanced protection, and long-term operational stability. From laboratory setups to defense and aerospace deployments, CRYONANO chargers ensure uninterrupted power where uptime and durability are essential.

KEY FEATURES

● Multi-Chemistry Charging Support

Compatible with lead-acid, lithium-ion, NiMH, NiCd, and custom battery chemistries using adaptive charging profiles for optimized battery life.

● Intelligent Charging Control

Advanced multi-stage and lifecycle-aware charging algorithms optimize charge curves for enhanced safety and longevity.

● Integrated Protection System

Provides overcharge, over-current, short-circuit, reverse polarity, and thermal overload protection for safe operation under all conditions.

● High Electrical Efficiency

Efficient AC-DC conversion and charging stages minimize power loss and thermal generation while ensuring stable DC output.

Notes

- Designed for repeated thermal cycling and long-duration experiments
- Performance depends on vacuum conditions and system integration
- External PID controller required for temperature regulation
- Custom configurations available for specific experimental setups

APPLICATIONS

- Defense and aerospace systems
- Off-grid and remote power installations
- Industrial automation and backup systems
- Telecom infrastructure and field systems
- Laboratory and scientific equipment
- Battery energy storage and management systems



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SPECIFICATIONS

ELECTRICAL PERFORMANCE

Input Type AC mains (single / three-phase, configuration dependent)

Output Type Regulated DC charging output

Charging Modes Multi-stage intelligent charging

Efficiency High-efficiency design (configuration dependent)

BATTERY COMPATIBILITY

Supported Chemistries Lead-acid, Li-ion, NiMH, NiCd

Charging Control Adaptive multi-chemistry algorithms

Lifecycle Management Integrated charge optimization

PROTECTION & SAFETY

Overcharge Protection Integrated

Over-Current Protection Integrated

Short-Circuit Protection Integrated

Reverse Polarity Protection Integrated

Thermal Protection Integrated

MECHANICAL & ENVIRONMENTAL

Operating Temperature -55°C to $+105^{\circ}\text{C}$

Enclosure Sealed, ruggedized

Vibration Resistance High (application dependent)

Cooling Convection / forced (configuration dependent)

CONFIGURABILITY

Architecture Options COTS / Modified / Fully Custom

Form Factor Embedded / rack / standalone

Output Configuration Single / multi-channel charging

Customization Voltage, current, environment-specific

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